

# Financial Copilot Platform

Capabilities & Testing Guide — May 2026

Repos: [financial-copilot-api](#) · [financial-copilot-ai-orchestrator](#) · [financial-copilot-web](#)

## What exists today

| Layer        | Implemented                                      | Not built / no UI                                 |
|--------------|--------------------------------------------------|---------------------------------------------------|
| Web          | Dashboard, demo seed, streaming AI chat          | Imports, account forms, auth, extra nav pages     |
| API          | Dashboard, accounts, holdings, seed, CSV imports | Tax API, scenarios, reports, bond optimiser, auth |
| Orchestrator | Health, chat, stream, session memory, routing    | DB memory, Phase C summarization                  |

## Recommended test order

0. Stack up (ports 8080, 8000, 3000) → 1. Health → 2. Demo seed → 3. Dashboard UI → 4. AI basic → 5. AI memory/streaming → 6. Agent routing → 7. Accounts/holdings (API) → 8. CSV imports (API)

## 0. Prerequisites

| Service      | URL                   | Notes                      |
|--------------|-----------------------|----------------------------|
| API          | http://localhost:8080 | Spring Boot + PostgreSQL   |
| Orchestrator | http://localhost:8000 | GOOGLE_API_KEY in .env     |
| Web          | http://localhost:3000 | NEXT_PUBLIC_AI_URL → :8000 |

**Pass when:** All three URLs respond without connection errors.

## 1. Infrastructure & connectivity

| #   | Capability            | How to test              | Pass criteria |
|-----|-----------------------|--------------------------|---------------|
| 1.1 | API liveness          | GET /actuator/health     | status "UP"   |
| 1.2 | Orchestrator          | GET /health              | status "ok"   |
| 1.3 | API from orchestrator | GET /health/dependencies | status "ok"   |
| 1.4 | Web loads             | Open :3000               | Page renders  |
| 1.5 | Agents list           | GET /api/v1/agents       | Lists agents  |

### PowerShell smoke:

```
Invoke-RestMethod http://localhost:8080/actuator/health
Invoke-RestMethod http://localhost:8000/health
Invoke-RestMethod http://localhost:8000/health/dependencies
Invoke-RestMethod http://localhost:8000/api/v1/agents
```

## 2. Demo data (required first)

| #   | Capability | How to test                       | Pass criteria               |
|-----|------------|-----------------------------------|-----------------------------|
| 2.1 | Seed demo  | UI: "Seed Demo Data"              | Metrics + holdings populate |
| 2.2 | Reseed     | POST /api/v1/demo/seed?reset=true | No 412 on dashboard         |

**Without seed:** dashboard and AI fail (412 — demo user not seeded).

### 3. Dashboard (web UI)

| #   | Capability        | How to test           | Pass criteria                               |
|-----|-------------------|-----------------------|---------------------------------------------|
| 3.1 | KPI cards         | After seed            | Net worth, assets, liabilities, health /100 |
| 3.2 | Region chart      | Left pie              | Segments + legend %                         |
| 3.3 | Asset class chart | Right pie             | Segments + legend %                         |
| 3.4 | Holdings table    | Scroll table          | Symbols, qty, ZAR values                    |
| 3.5 | Errors            | Stop API, refresh     | Error banner + retry                        |
| 3.6 | API parity        | GET /api/v1/dashboard | Matches UI numbers                          |

**Placeholder nav:** Portfolio, Retirement, Bonds are disabled — no pages.

### 4. AI chat — basic (web UI)

| #   | Capability   | How to test                  | Pass criteria            |
|-----|--------------|------------------------------|--------------------------|
| 4.1 | Send message | Ask net worth (one sentence) | User + assistant bubbles |
| 4.2 | Streaming    | Watch reply                  | Token-by-token growth    |
| 4.3 | Real data    | Compare to dashboard         | Figures align            |
| 4.4 | Brevity      | Long question                | 2–4 short paragraphs     |
| 4.5 | Agent label  | After reply                  | e.g. portfolio_analysis  |
| 4.6 | Disclaimer   | Panel footer                 | Educational only text    |
| 4.7 | Suggestions  | Empty-state chips            | Sends prefilled question |

### 5. AI chat — session & streaming

| #   | Capability     | How to test              | Pass criteria               |
|-----|----------------|--------------------------|-----------------------------|
| 5.1 | Multi-turn     | Net worth → "shorter"    | Context-aware shorter reply |
| 5.2 | Refresh        | F5 mid-chat              | Thread restored             |
| 5.3 | New chat       | New chat button          | Empty thread, new session   |
| 5.4 | Session API    | PowerShell script below  | History ≥ 2 msgs            |
| 5.5 | Non-stream API | POST /api/v1/chat        | reply + active_agent        |
| 5.6 | Stream API     | POST /api/v1/chat/stream | token + done events         |

#### Session test (PowerShell):

```
$sid = [guid]::NewGuid().ToString()
Invoke-RestMethod http://localhost:8000/api/v1/chat -Method POST ...
Invoke-RestMethod "http://localhost:8000/api/v1/chat/session/$sid"
Second POST with message "shorter" and same session_id
```

**Note:** Server memory is in-process (MemorySaver). Restarting orchestrator clears server sessions.

## 6. AI agent routing

Keyword routing; portfolio context loaded from API for specialist routes.

| Agent              | Test prompt             | Pass criteria            |
|--------------------|-------------------------|--------------------------|
| portfolio_analysis | How is my allocation?   | Uses holdings/allocation |
| tax                | Explain CGT on my gains | Agent label tax          |
| bond_optimisation  | Pay extra on home loan? | Uses liabilities         |
| risk_analysis      | Concentration risk?     | Risk bullets             |
| recommendation     | What next for TFSA?     | 2–3 suggestions          |
| scenario           | What if rates rise 2%?  | What-if answer           |
| report_generation  | Wealth summary report   | Brief snapshot           |
| general            | Hello                   | Conversational           |

## 7. Accounts & holdings (API only)

| #   | Capability     | Endpoint                       | Pass                |
|-----|----------------|--------------------------------|---------------------|
| 7.1 | List accounts  | GET /api/v1/accounts           | Demo accounts array |
| 7.2 | Create account | POST /api/v1/accounts          | 201                 |
| 7.3 | List holdings  | GET .../accounts/{id}/holdings | Holdings list       |
| 7.4 | Add holding    | POST .../holdings              | 201                 |
| 7.5 | Dashboard sync | Refresh dashboard / AI         | New data visible    |

Swagger: <http://localhost:8080/swagger-ui.html>

## 8. CSV import pipeline (API only)

See [financial-copilot-api/docs/IMPORTS.md](#) for full workflow.

| #   | Step         | Endpoint / action         | Pass                 |
|-----|--------------|---------------------------|----------------------|
| 8.1 | Source types | GET /imports/source-types | Lists source types   |
| 8.2 | Preview      | POST /imports/preview     | Column map suggested |
| 8.3 | Upload       | POST /imports/upload      | Job REVIEW           |
| 8.4 | Records      | GET .../records           | VALID/ERROR rows     |
| 8.5 | Fix          | PATCH + revalidate        | Errors cleared       |
| 8.6 | Commit       | POST .../commit           | Data persisted       |
| 8.7 | Bond CSV     | BOND_STATEMENT sample     | Balance updated      |
| 8.8 | Profiles     | mapping-profiles GET/POST | Save/reuse maps      |
| 8.9 | Cancel       | POST .../cancel           | Job cancelled        |

Samples: [backend/portfolio-service/src/main/resources/samples/](#)

## Not implemented yet (skip for current release)

- Login / auth / multi-user
- Tax engine REST API (AI discusses tax in prose only)
- Monte Carlo / scenarios API
- Report PDF generation
- Bond optimiser API
- Phase C — long-thread summarization
- Web UI for imports, accounts, settings

## Gate checklist before Phase C or deploy

- All services healthy; /health/dependencies ok
- Demo seeded
- Dashboard metrics, charts, holdings correct
- Streaming chat + follow-up "shorter" + refresh + new chat
- At least portfolio + one other agent route
- API-only: accounts/holdings/imports if in scope

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